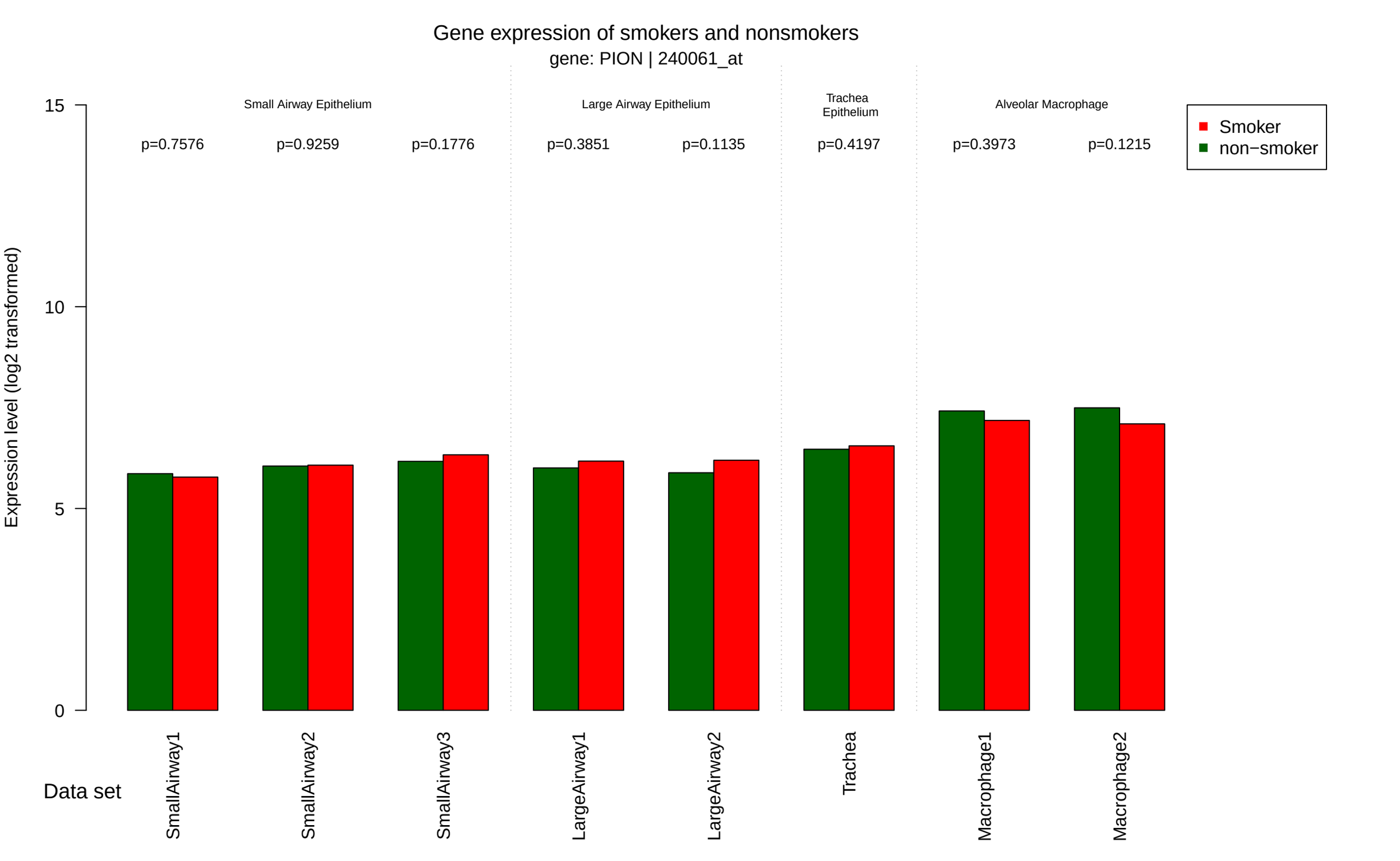
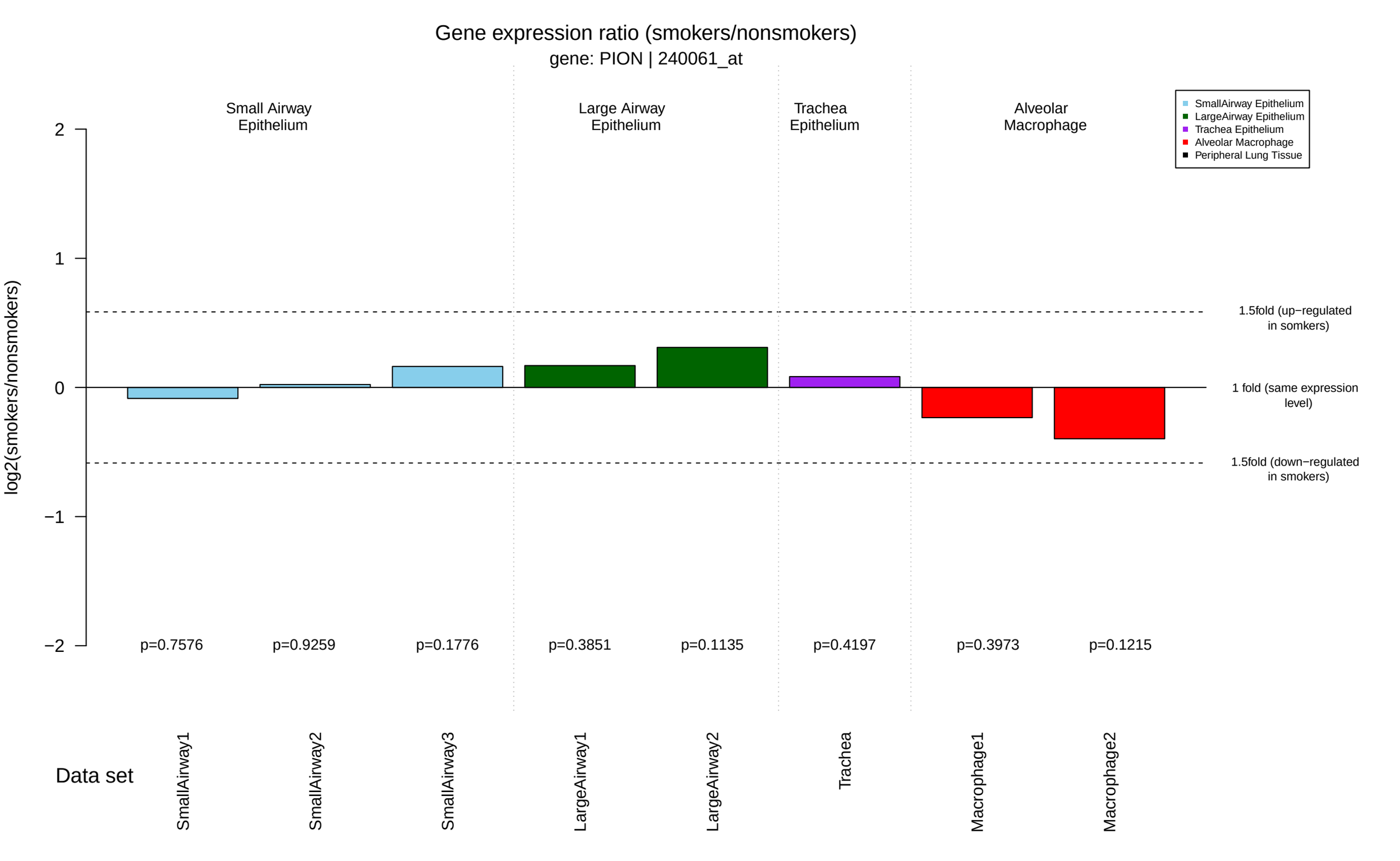
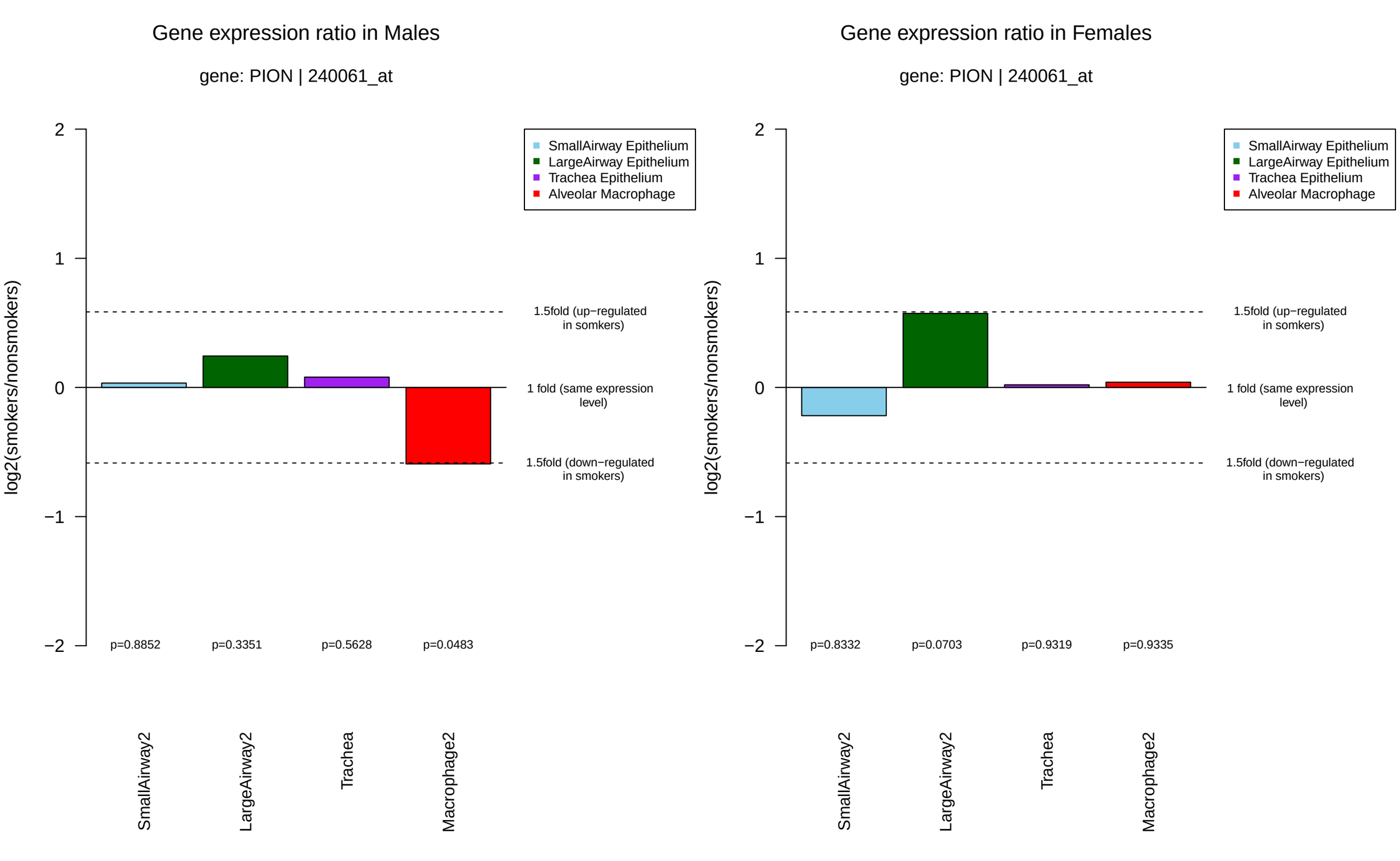
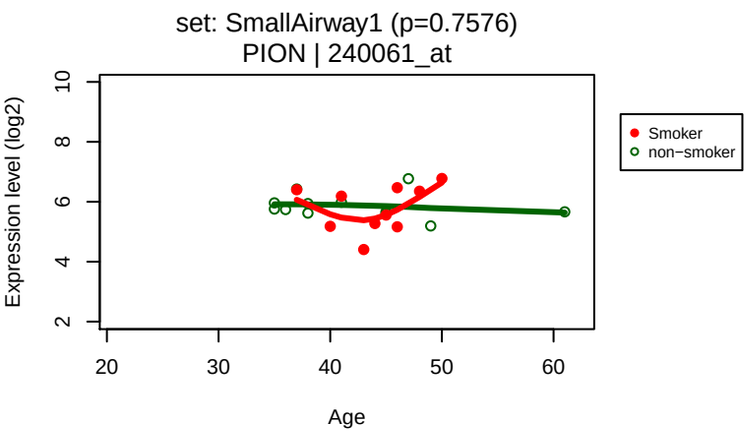
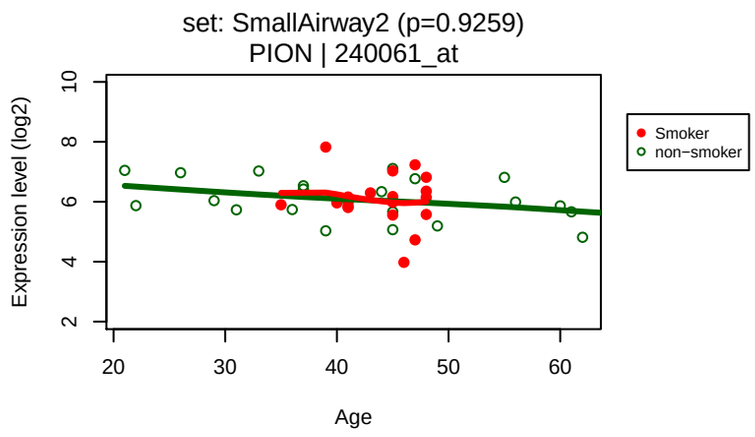
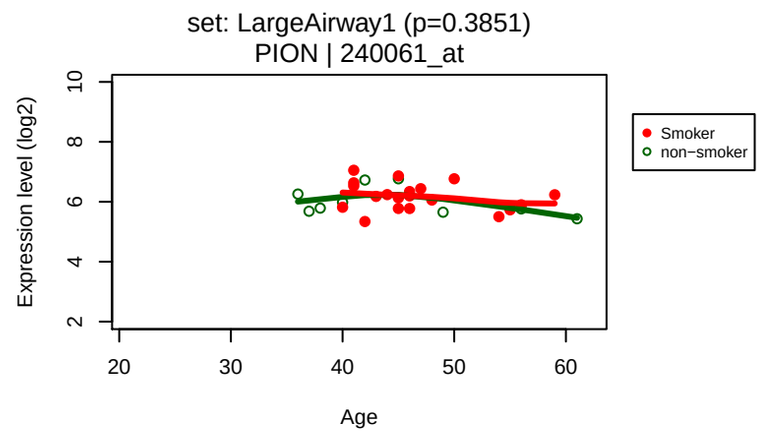

Gene expression of smokers and nonsmokers
gene: PION | 240061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7576
p=0.9259
p=0.1776
p=0.3851
p=0.1135
p=0.4197
p=0.3973
p=0.1215
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

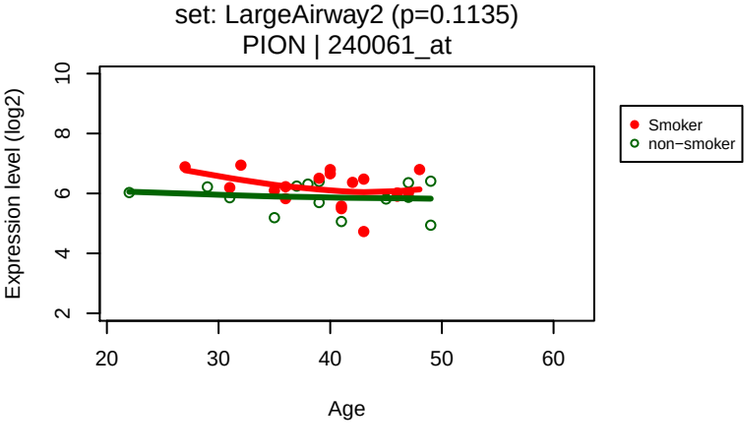

Gene expression ratio (smokers/nonsmokers)
gene: PION | 240061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7576
p=0.9259
p=0.1776
p=0.3851
p=0.1135
p=0.4197
p=0.3973
p=0.1215
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

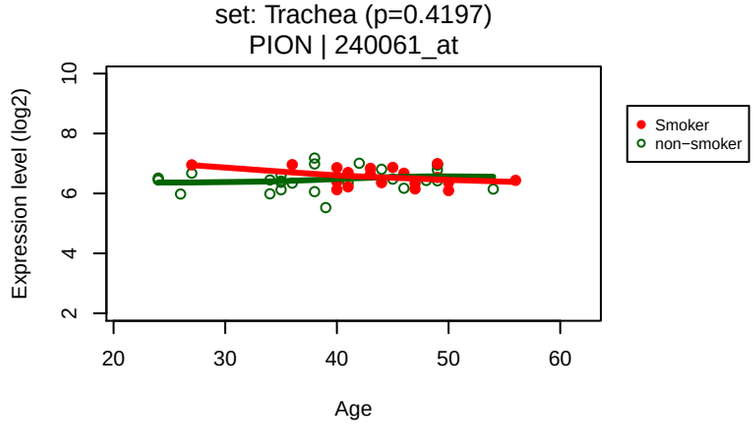

Gene expression ratio in Males
gene: PION | 240061_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8852
p=0.3351
p=0.5628
p=0.0483
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PION | 240061_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8332
p=0.0703
p=0.9319
p=0.9335
SmallAirway2
LargeAirway2
Trachea
Macrophage2

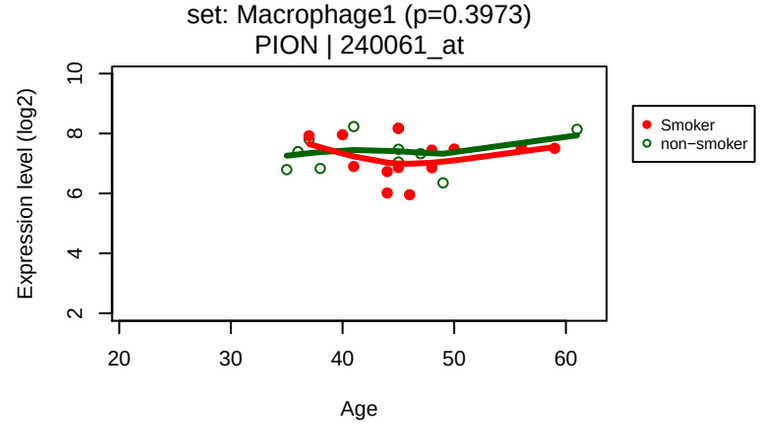

set: SmallAirway1 (p=0.7576)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker

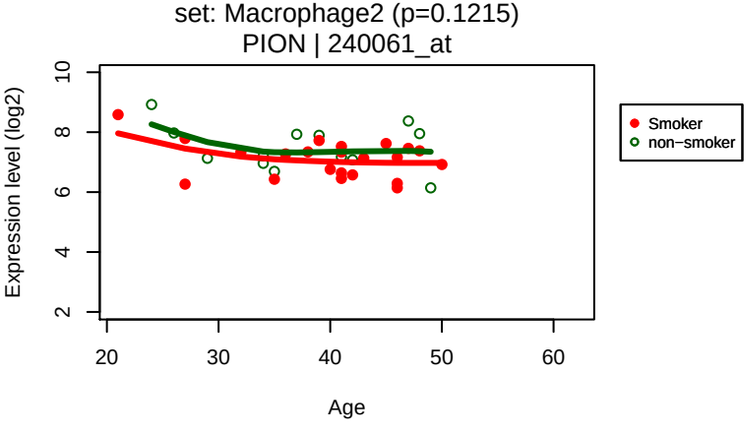

set: SmallAirway2 (p=0.9259)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker

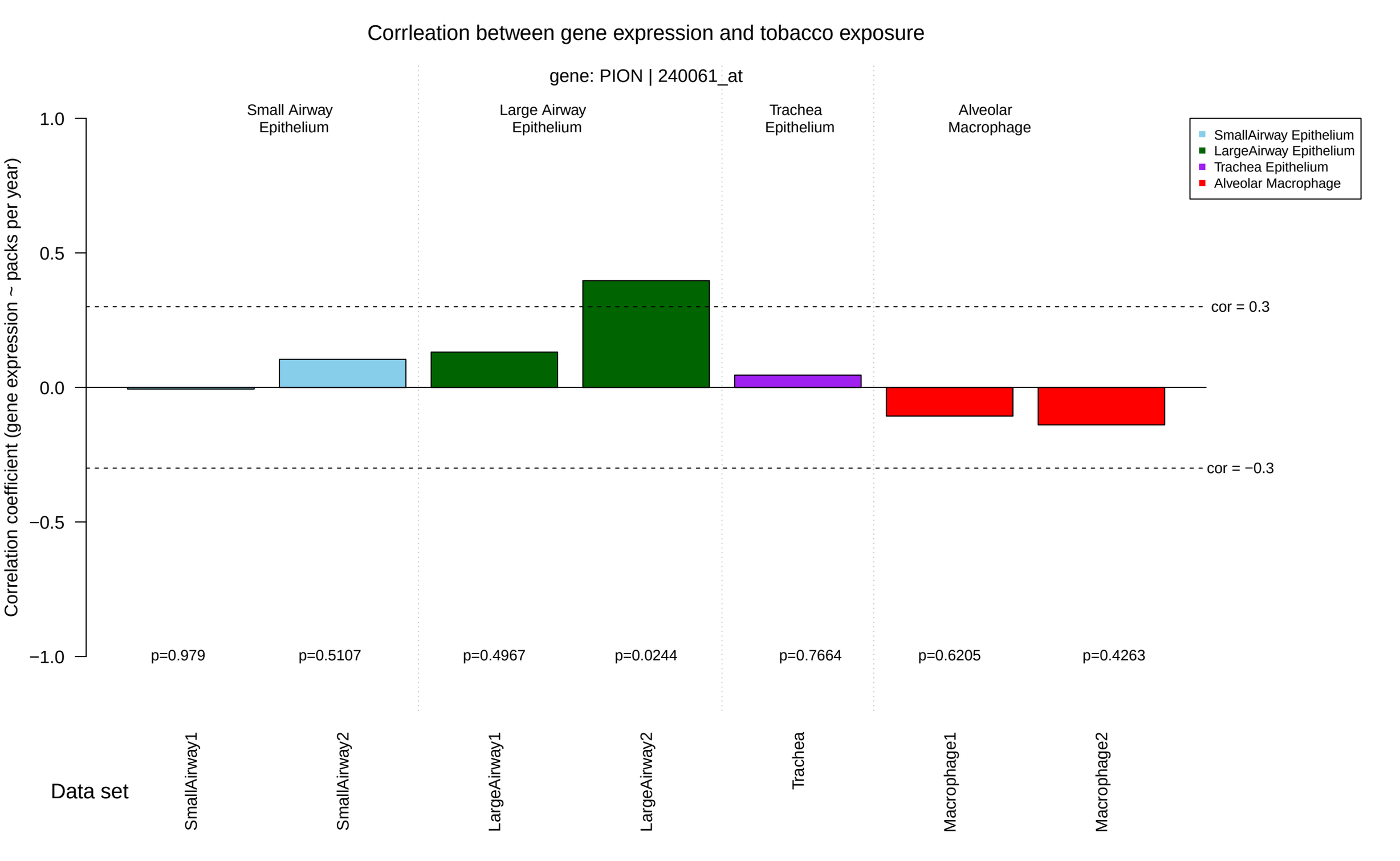

set: LargeAirway1 (p=0.3851)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1135)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4197)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3973)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1215)
PION | 240061_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PION | 240061_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.979
p=0.5107
p=0.4967
p=0.0244
p=0.7664
p=0.6205
p=0.4263
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2